

applications, and its flexibility has made it particularly useful for research and development. Even in high performance computing, where OpenStack was initially passed over due to its impact on performance, new applications that make use of well-managed OpenStack clouds are starting to emerge.

The challenges associated with OpenStack

The difficulties encountered in deploying and managing OpenStack clouds are well known. Highlighting some of these challenges helps to emphasise just how important management is to OpenStack's success.

Deployment is the most difficult stage of an OpenStack cloud. In theory, one can deploy OpenStack manually with no additional tools, but in practice, this requires a level of software knowledge and understanding that the majority of users lack.

OpenStack is highly evolved, with a number of different components, and OpenStack clouds must be installed on a functioning cluster. Before actually installing OpenStack, individual users must set up their cluster with the necessary hardware, operating system and a management method. Users must then determine which OpenStack components they need to install and configure.

The complex, multi-component nature of OpenStack is a challenge because the components interact in ways that often seem mysterious to users, especially those with limited OpenStack experience. This can make troubleshooting difficult. For example, if something is running more slowly than expected, it can be difficult to determine why, especially if the issues result from the complex interactions of the different components, or from a layer other than OpenStack itself. It is important to emphasise that with most approaches, OpenStack is only capable of managing itself. Troubleshooting the underlying hardware and software takes more work – or a more sophisticated management tool.

Enterprise management tools alleviate pain points

Given the challenges of deploying, managing and monitoring OpenStack, it is not surprising that a market has grown for enterprise management software to help users with these tasks. Most OpenStack management solutions focus primarily on deployment, though some assist in management and monitoring, and others are focused on management and monitoring of both OpenStack and the surrounding environment.

One example of the latter is Bright Computing's OpenStack management solution, which is based on stable and production-ready technology that manages the supercomputers at Boeing, Novartis, the US Department of Defense (DoD), the US Treasury, Lockheed Martin, and more than 500 other large enterprises around the world. This is in

contrast to some of the other solutions on the market, which are based on unproven technology.

For deployment, one of the most valuable tools offered is a comprehensive and intuitive installation wizard. For example, the Bright OpenStack features a wizard that poses high-level questions on networking and storage configuration, among other topics. The answers are then used to determine the configuration of the OpenStack deployment, ensuring that the final deployment meets the needs of users.

A drawback of many management solutions is that their deployment tools assume the user has already deployed the underlying cluster. Bright Computing's solution, however, can deploy onto bare metal hardware, which means it can handle both cluster and OpenStack deployment. This ensures that users can deploy their clouds successfully even if they lack the experience needed to deploy their cluster from bare metal hardware.

Given the complexity of OpenStack, the most useful solutions are those that focus on entire life cycle management, including the management of the underlying infrastructure and the OpenStack cloud software. The Bright OpenStack management solution offers multiple levels of management within the same solution. This allows users to manage the entire stack using a single interface comprising hardware, the operating system, cluster software and OpenStack. This single-pane-of-glass approach enables cloud admins to easily redistribute the underlying physical hardware among different users, tenants, instances and applications, adding or reconfiguring nodes, as needed.

This approach also makes it easier to monitor and check the health of each aspect of the stack. When an issue arises, such as slow runtime, the unified monitoring framework makes it easy to visualise issues by correlating different events in different layers of the entire stack. The single-pane-of-glass approach also improves the general usability of the cloud and its underlying infrastructure, putting all aspects of management directly into users' hands with the intuitive user interface.

High-quality management software makes it easy to turn bare metal hardware into a functional cluster with a healthy OpenStack cloud, something that any OpenStack user knows is difficult with OpenStack alone. Such software eliminates the headaches, helping users to capitalise on the value of OpenStack as an open source private cloud option – even without experienced cluster and cloud managers. 

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